

Food-Chain Dynamics in a Chemostat



SOICT

Hanoi University of
Science and
Technology
June 2026

Stability, bifurcation, and distinct removal
rates

Mathematical Modelling Project
School of Information and Communication Technology

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Question and Approach

Project question

Under what removal and growth conditions do prey and predators persist, wash out, coexist, or oscillate in a continuously operated chemostat?

What we reproduce

- ▶ A nutrient–prey–predator–top-predator model.
- ▶ Four equilibrium regimes, P_0 through P_3 .
- ▶ Stability, persistence, and near-Hopf behavior.

Why continuous-time ODEs

- ▶ Concentrations evolve continuously in a well-mixed vessel.
- ▶ Balance laws map directly to inflow, washout, growth, and predation.
- ▶ Analysis and simulation use the same state equations.

How the Chemostat Works



Core assumptions

- ▶ The vessel is continuously fed and perfectly mixed.
- ▶ Each organism consumes only the level immediately below it.
- ▶ Growth responses are increasing and saturating.
- ▶ Initial concentrations are nonnegative.

Why removal rates matter

- ▶ D_i combines dilution washout and organism-specific death.
- ▶ Distinct D_1, D_2, D_3 break the usual conservation law.
- ▶ Survival depends on growth exceeding removal at every trophic level.

The Dimensionless Model

$$\begin{aligned}\dot{S} &= 1 - S - f_1(S)x, \\ \dot{x} &= x(f_1(S) - D_1) - f_2(x)y, \\ \dot{y} &= y(f_2(x) - D_2) - f_3(y)z, \\ \dot{z} &= z(f_3(y) - D_3).\end{aligned}$$

Normalization

- ▶ Nutrient input: 1
- ▶ Nutrient dilution: $-S$
- ▶ Time scaled by dilution
- ▶ State $X = (S, x, y, z)^T$

Interpretation

- ▶ Positive terms create biomass.
- ▶ D_i removes biomass.
- ▶ Predation transfers biomass upward.

Monod Growth and Parameters

$$f_i(u) = \frac{a_i u}{k_i + u}, \quad f'_i(u) = \frac{a_i k_i}{(k_i + u)^2}$$
$$f_i^{-1}(d) = \frac{d k_i}{a_i - d}, \quad 0 \leq d < a_i.$$

Why Monod functions?

They are increasing, vanish without food, saturate at a_i , and preserve the paper's assumptions while giving reproducible formulas.

Parameter meanings

a_i : maximum growth rate

k_i : half-saturation level

D_1 : prey removal rate

D_2 : first-predator removal rate

D_3 : top-predator removal rate

Simulation input

$(a_1, k_1, a_2, k_2, a_3, k_3, D_1, D_2, D_3)$

Feasible Dynamics

Nonnegative invariant region

- ▶ At $S = 0$, $\dot{S} = 1 > 0$.
- ▶ At $x = 0$, $y = 0$, or $z = 0$, the corresponding derivative is zero.
- ▶ A positive trajectory cannot cross a coordinate plane.

Total concentration

$$T = S + x + y + z,$$

$$\dot{T} = 1 - S - D_1x - D_2y - D_3z.$$

Let

$$D_{\min} = \min(1, D_1, D_2, D_3),$$

$$D_{\max} = \max(1, D_1, D_2, D_3).$$

Then

$$1 - D_{\max}T \leq \dot{T} \leq 1 - D_{\min}T.$$

Consequence

Solutions stay meaningful and eventually enter a common bounded region. The model is dissipative.

Four Equilibria, Four Regimes

State	Positive populations	Defining relations	Biological regime
P_0	none	$(1, 0, 0, 0)$	Complete washout
P_1	x	$f_1(S_1) = D_1$	Prey only
P_2	x, y	$f_2(x_2) = D_2$	Top predator absent
P_3	x, y, z	$f_3(y_3) = D_3$	Full coexistence

$$S_1 = f_1^{-1}(D_1), \quad x_1 = \frac{1 - S_1}{D_1} \qquad x_2 = f_2^{-1}(D_2), \quad y_3 = f_3^{-1}(D_3)$$

P_3 requires a coupled numerical solve for S_3 and x_3 .

Invasion Controls Stability

Per-capita invasion rates

$$\lambda_x = f_1(1) - D_1,$$

$$\lambda_y = f_2(x_1) - D_2,$$

$$\lambda_z = f_3(y_2) - D_3.$$

Positive invasion creates the next trophic regime;
negative invasion stabilizes the lower-level
boundary equilibrium.

Equilibrium	Main stability test
P_0	$\lambda_x < 0$
P_1	$\lambda_y < 0$
P_2	cubic RH and $\lambda_z < 0$
P_3	quartic RH test

Local stability

Every eigenvalue of the Jacobian
must have negative real part.

Persistence, Hopf, and Model Choice

Uniform persistence

- ▶ Dissipativity prevents unbounded escape.
- ▶ Boundary equilibria must repel each missing predator.
- ▶ Under the paper's extra assumptions, every positive component stays bounded away from zero asymptotically.

$$\liminf_{t \rightarrow \infty} (S, x, y, z) \geq \eta > 0.$$

Implementation choice: printed equation (1.2) conflicts with the later Jacobian, so the code follows the internally consistent analytical form.

Hopf boundaries

$$P_2 : \quad A_1 A_2 = A_3$$

$$P_3 : \quad C_1 C_2 C_3 = C_3^2 + C_1^2 C_4$$

A conjugate eigenvalue pair must cross the imaginary axis with nonzero speed; damped oscillation alone is not a proof of bifurcation.

Software Architecture

scenario_parameters.csv $\longrightarrow$ model + equilibria $\longrightarrow$ Euler
/ RK4 $\longrightarrow$ diagnostics + figures

Subsystem	Responsibility
equilibria_stability/ simulator/ main_simulations/ sensitivity + robustness demo/	Monod functions, P_0 – P_3 , positivity checks RHS, Euler, RK4, CSV output, diagnostics Five biological regimes and comparison plots Parameter sweeps, elasticity, numerical robustness Interactive equations, thresholds, and trajectories

Euler and RK4

Forward Euler

$$X_{n+1} = X_n + hF(X_n)$$

- Transparent first-order baseline

Classical Runge–Kutta

$$k_1 = F(X_n),$$

$$k_2 = F(X_n + \frac{h}{2}k_1),$$

$$k_3 = F(X_n + \frac{h}{2}k_2),$$

$$k_4 = F(X_n + hk_3),$$

$$X_{n+1} = X_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

- Fourth-order method for long, oscillatory runs
- Partial step reaches t_{end} exactly

Validation Before Interpretation

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automated
tests passed

Test area	Evidence
Model RHS	Known derivative, no mutation, inward boundary field, equilibria are stationary
Solvers	Euler recurrence, RK4 accuracy/order, exact final time, invalid-input rejection
Safeguards	Negative clamp count, explosion stop, finite-state checks
Diagnostics	Tail convergence, min/max, equilibrium distance, CSV schema

Production run: all five RK4 scenarios completed with zero negative clamps and no explosion; every equilibrium scenario matched its target.

Five Numerical Experiments

Scenario	Target	Expected long-run behavior
Washout	P_0	Nutrient returns to 1; all organisms vanish
Prey only	P_1	Prey persists; both predators vanish
Two-level survival	P_2	Prey and first predator persist
Coexistence	P_3	All four state variables remain positive
Oscillatory response	P_3	Slowly damped oscillation near coexistence

Method

RK4, $h = 0.01$

Initial state

$(0.5, 0.2, 0.1, 0.05)$

Horizon

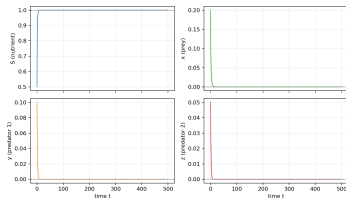
500; oscillatory run 2000

Source: Fixed representative parameter sets; no random generation and no experimental calibration.

The Washout Hierarchy

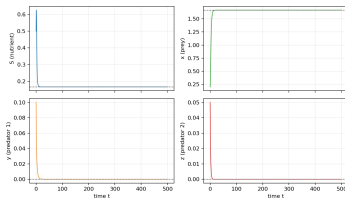
P_0 : complete washout

washout_P0: Complete organism washout
(dashed line: theoretical equilibrium)



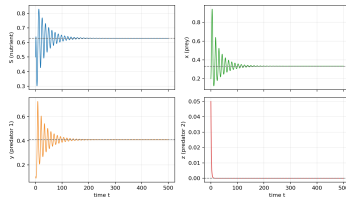
P_1 : prey survives

prey_only_P1: Prey survives; both predators wash out
(dashed line: theoretical equilibrium)



P_2 : two levels survive

prey_first_predator_P2: Prey and first predator survive
(dashed line: theoretical equilibrium)



$S \rightarrow 1$; distance 5.55×10^{-15}

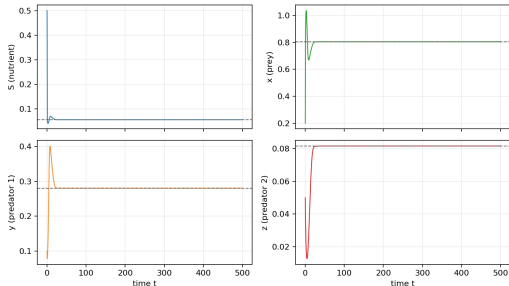
$(S, x) \rightarrow (0.1667, 1.6667)$

Distance 1.29×10^{-6}

Each upper trophic level persists only when its invasion growth exceeds its removal rate.

Full Coexistence at P_3

coexistence_P3: All trophic levels coexist
(dashed line: theoretical equilibrium)



Theoretical equilibrium

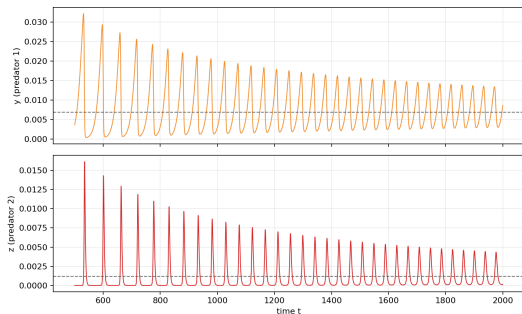
$$\begin{pmatrix} 0.055067 \\ 0.804597 \\ 0.280534 \\ 0.081550 \end{pmatrix}$$

Numerical distance

$$1.94 \times 10^{-11}$$

Slowly Damped Oscillations

Oscillatory scenario: slowly decreasing predator-cycle amplitude



Long-run evidence

- After $t = 500$: 23 peaks in y , 20 in z
- Tail-mean distance to P_3 : 4.66×10^{-4}
- Late/middle amplitude ratio: 0.425

Conclusion

Damping is consistent with stable coexistence near a Hopf boundary.

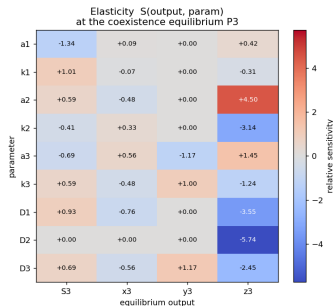
Theory Predicts the Outcome

Scenario	Theoretical target	Nearest state	Distance
Washout	P_0	P_0	5.55×10^{-15}
Prey only	P_1	P_1	2.57×10^{-14}
Two-level survival	P_2	P_2	1.29×10^{-6}
Coexistence	P_3	P_3	1.94×10^{-11}
Oscillatory response	stable P_3 nearby	P_3	2.49×10^{-3} at final time

What simulation demonstrates

Numerical trajectories reproduce the equilibrium hierarchy and provide evidence consistent with the analytical stability thresholds.

Removal Rates Change the Regime



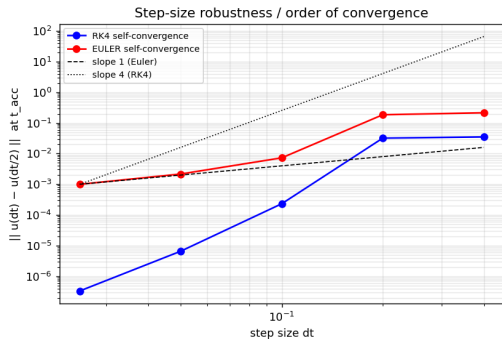
Coexistence lost near

Sweep	Boundary	Transition
D_1	1.0	$P_3 \rightarrow P_2$
D_2	1.4	$P_3 \rightarrow P_2$
D_3	0.925	$P_3 \rightarrow P_2$

Largest local effects on z_3

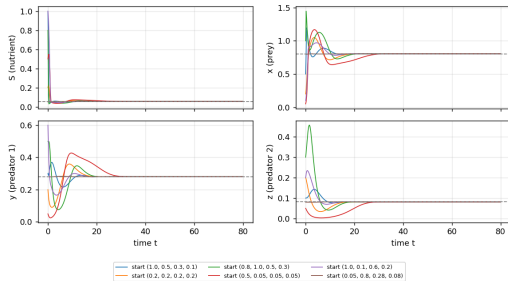
- D_2 : elasticity -5.742
- a_2 : elasticity $+4.501$
- D_1 : elasticity -3.552

The Model Is Numerically Robust



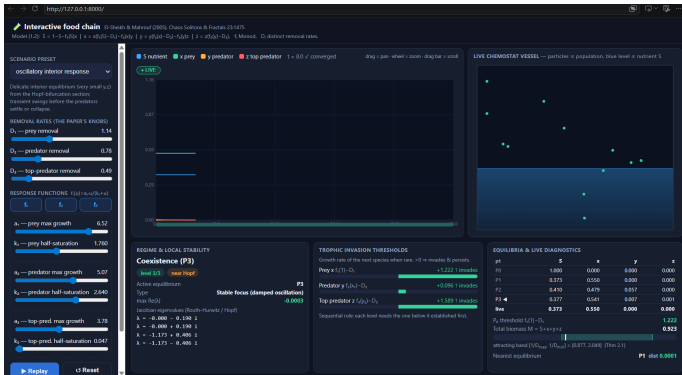
Step-size test RK4 converges; coarse Euler may select the wrong regime.

Initial-condition robustness (coexistence P_3):
all starts reach the same equilibrium (dashed line)



Initial-condition test Six positive starts reached P_3 within 1.95×10^{-11} .

Interactive and Reproducible



Run the simulator `python3 -m simulator.runner`

`-scenario coexistence_P3`

Run every experiment
`python3 -m`

`main_simulations.run_all`

`python3`

`robustness_sensitivity/`

`run_all.py`

Start the browser demo

`python3 -m demo.app`

What the Project Establishes

Reference conclusions

- ▶ Distinct removal rates remove the simple conservation law.
- ▶ Solutions are nonnegative, bounded, and dissipative; invasion thresholds organize washout and coexistence.
- ▶ Persistence can hold and Hopf bifurcation can create cycles.

Our numerical conclusions

The simulator reproduces P_0 – P_3 , damped oscillations, and removal-driven regime changes with robust RK4 results.

Limitations

- ▶ Parameters are constructed examples, not experimental estimates.
- ▶ The project illustrates but does not continue a Hopf branch.
- ▶ Initial-condition tests cannot prove global persistence, and calibration data are unavailable.

Central lesson

Every trophic level must grow fast enough on its equilibrium food supply to overcome its own removal.

Questions?

Stability, simulation, sensitivity, or biological interpretation

Code and reproducible outputs are available in the project repository.

Backup: Complete Parameter Sets

Scenario	a_1	k_1	a_2	k_2	a_3	k_3
P_0 washout	0.60	0.50	1.50	0.50	1.20	0.40
P_1 prey only	2.00	0.50	0.50	1.00	1.00	1.00
P_2 two levels	2.00	0.50	2.00	1.00	0.50	1.00
P_3 coexistence	4.80	0.17	4.30	1.85	4.60	1.66
Oscillatory	6.52	1.76	5.07	2.64	3.78	0.0466

Scenario	D_1	D_2	D_3	Horizon
P_0 washout	0.80	0.70	0.70	500
P_1 prey only	0.50	0.60	0.60	500
P_2 two levels	0.50	0.50	0.70	500
P_3 coexistence	0.72	1.11	0.665	500
Oscillatory	1.14	0.778	0.488	2000

Backup: Equilibrium Formulas

$$P_0 = (1, 0, 0, 0),$$

$$P_1 = (S_1, x_1, 0, 0), \quad S_1 = f_1^{-1}(D_1), \quad x_1 = \frac{1 - S_1}{D_1},$$

$$P_2 = (S_2, x_2, y_2, 0), \quad x_2 = f_2^{-1}(D_2),$$

$$S_2 + f_1(S_2)x_2 = 1, \quad y_2 = \frac{x_2(f_1(S_2) - D_1)}{D_2},$$

$$P_3 = (S_3, x_3, y_3, z_3), \quad y_3 = f_3^{-1}(D_3),$$

$$S_3 + f_1(S_3)x_3 = 1, \quad x_3(f_1(S_3) - D_1) = f_2(x_3)y_3,$$

$$z_3 = \frac{y_3(f_2(x_3) - D_2)}{D_3}.$$

Backup: Jacobian and Routh–Hurwitz

$$J = \begin{pmatrix} -1 - f'_1(S)x & -f_1(S) & 0 & 0 \\ f'_1(S)x & f_1(S) - D_1 - f'_2(x)y & -f_2(x) & 0 \\ 0 & f'_2(x)y & f_2(x) - D_2 - f'_3(y)z & -f_3(y) \\ 0 & 0 & f'_3(y)z & f_3(y) - D_3 \end{pmatrix}.$$

P_2 **cubic**

$$\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3$$

$$A_1, A_2, A_3 > 0, \quad A_1A_2 > A_3,$$

together with $f_3(y_2) - D_3 < 0$.

P_3 **quartic**

$$\lambda^4 + C_1\lambda^3 + C_2\lambda^2 + C_3\lambda + C_4$$

$$C_i > 0, \quad C_1C_2 > C_3,$$

$$C_1C_2C_3 > C_3^2 + C_1^2C_4.$$

Backup: Reproduction and References

Commands `python3 -m unittest discover`

`-v`

`python3 -m main_simulations.run_all`

`python3 robustness_sensitivity/run_all.py`

`python3 -m demo.app`

Trajectory schema

t, S, x, y, z

Fixed scenario input:

$a_1, k_1, a_2, k_2, a_3, k_3, D_1, D_2, D_3$

Primary reference

M. M. A. El-Sheikh and S. A. A. Mahrouf,
“Stability and bifurcation of a simple food chain
in a chemostat with removal rates,” *Chaos,
Solitons and Fractals*, 23 (2005), 1475–1489.
DOI: 10.1016/j.chaos.2004.06.079.

Project evidence

- ▶ Source code and automated tests
- ▶ Generated CSV and JSON diagnostics
- ▶ Simulation, sensitivity, robustness, and course requirements